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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS



UPES UNIVERSITY OF TOMORROW

# Elements of AIML

## Unit 03 – Lecture 01: Introduction to Machine Learning & Datasets

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February 28, 2026

Topic focus: **what it means to learn from data** + dataset basics

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

# Quick Links

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# Agenda

ML basics

Datasets

Feature sets

Mini example

Summary

# Learning Outcomes

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By the end of this lecture, you should be able to:

- Define **machine learning** and describe common task types.
- Explain what a **dataset** is (rows, columns, features, labels).
- Distinguish **features** vs **labels** and list common data types.
- Identify basic data issues (missing values, outliers, noise) that affect ML.

# Abbreviations & Symbols

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- $n$: number of examples (rows)
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- $f(\cdot)$: model/function learned from data

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# What is Machine Learning?

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ML is about building models that **learn patterns from data** to make predictions or decisions.

- Instead of writing rules manually, we learn a function from examples.
- Learning requires: data, an objective, and a way to evaluate performance.

# A Standard Definition (Mitchell)

A program learns from **experience** $E$ with respect to some **task** $T$ and **performance measure** $P$ if its performance at $T$, as measured by $P$, improves with $E$.

## Example (spam detection)

$T$: classify email as spam/ham, $E$: labeled emails, $P$: accuracy/F1 on unseen emails.

# ML Workflow (Big Picture)

Data → Prep / EDA → Train Model → Evaluate

Key idea: evaluation must reflect the **future/unseen** setting.

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- **Supervised learning**: learn from labeled data
  - regression (predict a number), classification (predict a class)
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  - clustering, dimensionality reduction
- **Reinforcement learning**: learn via rewards from interaction (Unit IV overview)

# Tasks at a Glance (What changes?)

| Task | Input | Output |
|---|---|---|
| Regression | $x$ features | number (e.g., price) |
| Classification | $x$ features | class (e.g., spam/ham) |
| Clustering | $x$ only | group/cluster id |
| Dim. reduction | $x$ only | lower-dim representation $z$ |

Same dataset can support multiple tasks depending on the objective.

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- Many ML problems use a table: **rows** are examples, **columns** are variables.
- Inputs are **features** ($X$), output is **label/target** ($y$).
- Goal: learn a mapping $f : X \to y$ that generalizes.

# Dataset Notation (Typical)

- Feature matrix:

$$X \in \mathbb{R}^{n \times d}$$

$n$ = samples, $d$ = features, $K$ = classes.

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- Classification target:

$$y \in \{1, \ldots, K\}^{n}$$

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- **Numerical**: integers, floats (e.g., age, income)
- **Categorical**: finite set of values (e.g., city, department)
- **Ordinal**: ordered categories (e.g., low/medium/high)
- **Text / images / time series**: require feature representations

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- Good features make learning easier; bad features can make any model fail.
- Feature sets often require:
  - encoding (e.g., one-hot), scaling, cleaning, transformations

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- City name (categorical) → one-hot encoding
- Time stamp → hour-of-day, day-of-week, seasonality features
- Image → pixel values or learned features (CNN)

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# Common Data Problems (Preview to Labs)

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These issues are the focus of the lab experiments on EDA and preprocessing.

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- Missing values (NaN): require imputation or removal
- Outliers: can distort averages and some models
- Duplicates and inconsistent formatting
- Noisy labels: incorrect ground truth

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# Example: House Price Prediction

- Goal: predict price (target $y$)

Question: what if **location** is categorical? $\rightarrow$ we must encode it.

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# Example: House Price Prediction

- Goal: predict price (target $y$)
- Features ($X$): area, number of rooms, location, age of house
- Regression model learns: $Price \approx f(Area, Rooms, Location, Age)$

Question: what if **location** is categorical? $\rightarrow$ we must encode it.

# Mini Worked Example (Tiny Dataset)

| Location | Area | Rooms | Price |
|---|---|---|---|
| A | 1200 | 2 | 55 |
| B | 1500 | 3 | 72 |
| A | 900 | 2 | 48 |

To use **Location**, encode it (e.g., one-hot) instead of treating A/B as numeric.

# Summary

- ML learns a mapping from data; supervised vs unsupervised vs RL.
- Datasets contain examples (rows) and variables (columns).
- Features are representations; data quality strongly affects results.
- Next: train/test/validation splits and cross-validation (Lecture 02).

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

# Exit Questions

1. Define machine learning in your own words.
2. What is the difference between features and labels?
3. Give two examples each of supervised and unsupervised tasks.
4. Name three common dataset issues that affect model performance.